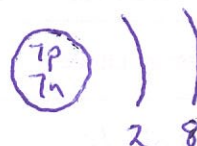
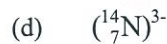
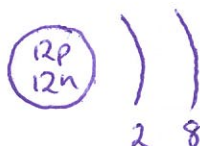
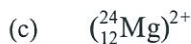
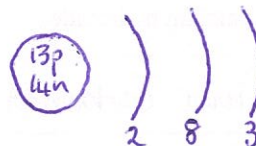
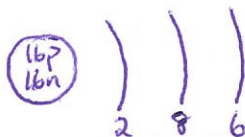


**YEAR 10 SCIENCE EXTENSION
NATURAL AND PROCESSED MATERIALS
ASSIGNMENT 1: ATOMIC STRUCTURE AND EQUATIONS**

NAME: SOLUTIONSTotal: 54

1. Draw the electron shell configuration for each of the following.



(1½ marks each)

(6)

2. Using $^{35}_{17}\text{Cl}^{1-}$ as an example, explain the following.(a) Were electrons gained or lost to form the Cl^{1-} ion?

gained.

(1)

(b) How many electrons were gained or lost?

1

(1)

(c) In which group is Cl?

VII

(1)

(d) What valency do all members of this group possess?

1-

(1)

(e) Explain carefully why the $^{35}_{17}\text{Cl}^{1-}$ ion is stable, referring to the Octet Rule.

- Cl atom has 7 electrons in the outer shell. (1)

- It accepts 1 electron. (1)

- This gives the same electron configuration as the nearest inert gas (Ar), which is very stable. (1)

(3)

3. Complete the following table.

NAME	FORMULA
carbon dioxide	CO_2
aluminium sulphate	$\text{Al}_2(\text{SO}_4)_3$
dinitrogen tetroxide	N_2O_4
calcium hydroxide	$\text{Ca}(\text{OH})_2$
ammonium carbonate	$(\text{NH}_4)_2\text{CO}_3$
copper (II) chloride	CuCl_2
nitrogen dioxide	NO_2
potassium nitrate	KNO_3
zinc sulphide	ZnS
sodium phosphate	Na_3PO_4

($\frac{1}{2}$ mark each)

(5)

4. Complete the following table.

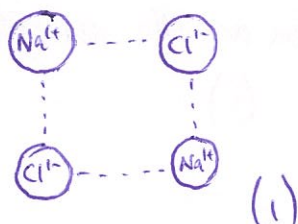
ELEMENT/ION	PROTON NUMBER	NEUTRON NUMBER	ELECTRON NUMBER
$^{24}_{12}\text{Mg}$	12	12	12
$^{35}_{17}\text{Cl}^{1-}$	17	18	18
$^{56}_{26}\text{Fe}$	26	30	26
$^{27}_{13}\text{Al}^{3+}$	13	14	10
$^{16}_8\text{O}$	8	8	8

($\frac{1}{2}$ mark off each mistake)

(5)

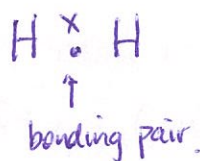
5. Using clear and simple diagrams to help, explain how the following bonds arise.

(a) ionic



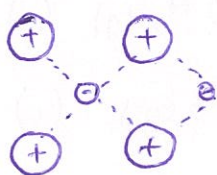
- Electrostatic attraction between oppositely-charged ions. (1)
- Ions are arranged in a lattice.

(b) covalent



- Attraction for a shared pair of electrons by 2 adjacent nuclei. (1)
- Bonding pair of electrons move in the region between the two H nuclei. (1)

(c) metallic

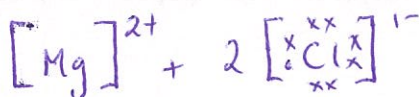


- Electrostatic attraction between metal ions and delocalised electrons. (1)
- The electrons move between the ions and maintains the metallic bond. (1)

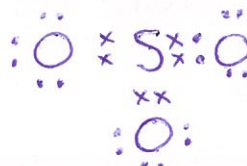
(6)

6. Draw **electron dot diagrams** for the following compounds.

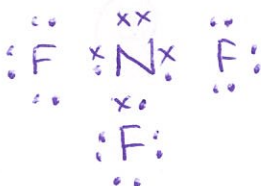
(a) MgCl_2



(b) SO_3



(c) NF_3



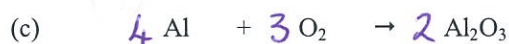
(d) K_2O



(1½ marks each)

(6)

7. Balance the following equations.

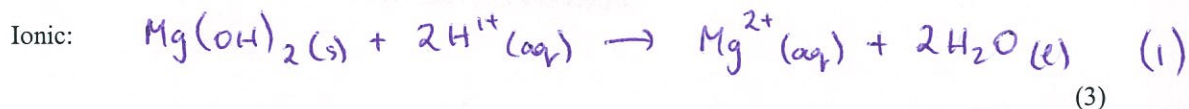
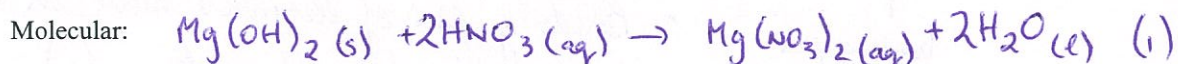


(1 mark each)

(5)

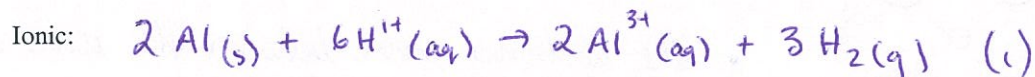
8. Write **complete word equations** for the following acid reactions. Underneath each, write a **balanced molecular equation** and a **balanced ionic equation**.

(a) magnesium hydroxide + nitric acid → magnesium nitrate + water (1)



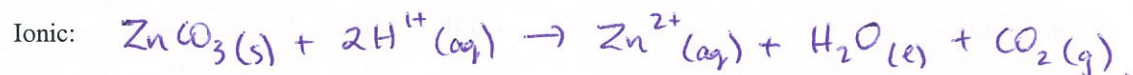
(3)

(b) aluminium + hydrochloric acid → aluminium chloride + hydrogen (1)



(3)

(c) zinc carbonate + nitric acid → zinc nitrate + water + carbon dioxide



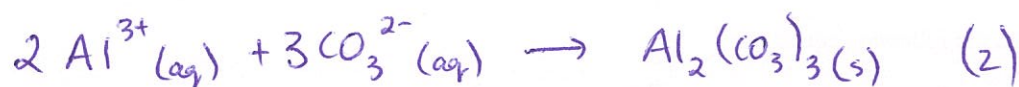
(3)

9. Predict whether a precipitate is produced when the following solutions are mixed together. For those that produce a precipitate, write a **balanced ionic equation**. Write "no reaction" if a precipitate is not produced.

(a) barium nitrate solution + potassium sulphate solution



(b) aluminium nitrate solution + sodium carbonate solution



(c) lead nitrate solution + copper (II) acetate solution

NO REACTION (1)

(5)